Supplementary Material for: One Checkpoint for Complete, Slide-Only and RNA-Only Patients: Auditing and Training Histology–Transcriptomics Survival Models for Missing Inputs

Sharoon Sharif

TABLE S1
EVALUATION PROTOCOL, IDENTICAL FOR ALL METHODS.

Aspect	Choice
Unit	patient; all slides of a patient concatenated
Splits	SurvPath's five-fold patient-level splits; the validation fold is the reported fold
Endpoints	disease-specific survival (primary); overall survival (secondary)
Time bins	quartiles of uncensored training patients, outer edges $\pm\infty$, reused for validation
Metrics	Harrell's C-index on continuous time [43], [44] (primary); Uno's IPCW C-index [45] truncated at the 75th percentile of training event times, integrated Brier score [46] on the interior quartile grid, log-rank test at the median risk split (secondary)
Budget, seeds	20 epochs, final checkpoint; seeds 0, 1, 2 with the fold index added
Statistics	one value per (cohort, fold) = mean over seeds; paired t and exact Wilcoxon tests over the 25 pooled folds, 95% percentile-bootstrap CI (10,000 resamples); per-cohort tests ($n = 5$) Holm-adjusted, exploratory
Audit	each input removed or permuted across patients; 10–50% of patients missing one modality at random (five draws); Spearman correlation of fused and single-input risks

S-I. PROTOCOL AND METHODS

"S"-prefixed sections, tables, figures and equations are in this document; unprefixed ones are in the main paper. In the tables, PQF denotes PathQ-Former and PQF$^-$ its variant without auxiliary heads. Table S1 states the evaluation protocol, identical for every method, and Table S2 the methods and seeds. SurvPath (official implementation, published hyper-parameters, 4,096 patches per slide, 21.2 M parameters), ABMIL, SNN and the two-layer RNA MLP use the hyper-parameters of the SurvPath repository; none was tuned under this protocol. PathQ-Former reads all patches of a patient (6,214 for the median of the 40 profiled BLCA fold-0 patients, Table S14) and has 16.9 M parameters including the pathway MLPs. Every results table (Section S-III here, Section IV of the main paper) is regenerated from the archived per-run files by `scripts/reproduce_tables.sh` in the code repository (https://github.com/SharoonSharif/PathQFormer; Zenodo DOI 10.5281/zenodo.22900123). Bracketed reference numbers refer to the reference list of the main paper.

S-II. TRAINING DETAILS AND SELECTION HISTORY

Query blocks. Each modality $m \in \{h, g\}$ has $K = 32$ learned queries $Q^m \in \mathbb{R}^{K\times d}$ with $d = 256$ and a stack of $L = 2$ pre-norm blocks,

$$Q \leftarrow Q + \mathrm{CA}\big(\mathrm{LN}(Q),\, U^m\big), \tag{S1}$$

$$Q \leftarrow Q + \mathrm{SA}\big(\mathrm{LN}(Q)\big), \qquad Q \leftarrow Q + \mathrm{FFN}\big(\mathrm{LN}(Q)\big), \tag{S2}$$

TABLE S2
METHODS AND SEEDS UNDER THE 20-EPOCH DSS PROTOCOL. ONE BRCA RUN OF EACH SINGLE-MODALITY PATHQ-FORMER IS MISSING AT SEED 2.

Method	Inputs	Seeds	Also run under
SurvPath (official)	WSI+RNA	3	OS (1); 10 epochs (3)
ABMIL	WSI	3	–
SNN	RNA	3	–
RNA MLP (two layers)	RNA	3	–
Slide-only PathQ-Former	WSI	3 (BRCA 2)	–
RNA-only PathQ-Former	RNA	3 (BRCA 2)	–
Late fusion	WSI+RNA	2	–
PQF$^-$	WSI+RNA	3	10 epochs (3); Table S12 (1)
PQF	WSI+RNA	3	OS (1); Table S13 (1)

TABLE S3
TRAINING CONFIGURATION OF PATHQ-FORMER AND OF THE OFFICIAL SURVPATH BASELINE. EQ. (1) IS THE DISCRETE-TIME NEGATIVE LOG-LIKELIHOOD OF THE MAIN PAPER.

Setting	PathQ-Former	SurvPath (official)
Epochs	20, final checkpoint	20, final checkpoint
Optimizer	AdamW [49], lr 10^{-4}, wd 10^{-4}, 1 warm-up epoch, cosine decay	RAdam [50], lr 5×10^{-4} (published), wd 10^{-4}
Batch	1 patient, gradient accumulation 2, gradient clip 1.0	1 patient
Dropout	0.25; modality dropout 0.15	0.1
Patches	all patches of all slides	4,096-patch subsample (published)
Loss	Eq. (1), censored term unweighted; auxiliary heads $\lambda = 0.5$ (full model) or 0	Eq. (1), censored term weighted 0.5 (published)
Sampling	uniform over training patients	weighted by (bin, censorship) (published)
Genes	min–max to $[-1, 1]$, fit on the training split	same
Parameters	16.9 M	21.2 M

where U^m are the modality's input tokens (the LayerNormed patch embeddings, or the 275 pathway tokens produced by per-pathway two-layer MLPs of hidden width 128), CA is 8-head cross-attention with queries from Q and keys and values from U^m, SA is self-attention among the queries and FFN a two-layer feed-forward network. Cross-attention over N input tokens costs $O(KNd)$ and attention among the queries $O(K^2 d)$; the block returns 32 vectors, so the fusion stage does not depend on N. The fused codes pass through two Transformer encoder layers ($d = 256$, 8 heads) and a learned-query attention pooling; a linear layer with a sigmoid produces the $T = 4$ hazards (Table S3).

Selection history. The base hyper-parameters of PathQ-Former (learning-rate schedule, dropout, modality-dropout rate, number of queries, fusion depth) were chosen on BLCA under the 10-epoch budget from the single-seed ablation runs in Table S12; none is distinguishable at one seed, and the chosen values are defaults rather than optima. The budget was raised from 10 to 20 epochs after inspecting the pooled validation curves of PathQ-Former and SurvPath under the 10-epoch budget; the 20-epoch curves (Fig. S1; the per-cohort curves, `epoch_curves_by_cohort.pdf`, are in the

Manuscript received October XX, 2026.

S. Sharif is with the Department of Electrical Engineering and Computer Science, Alabama A&M University, Normal, AL 35762 USA, and with Faith Comes By Hearing, Albuquerque, NM USA (e-mail: sharoonsharif@gmail.com).

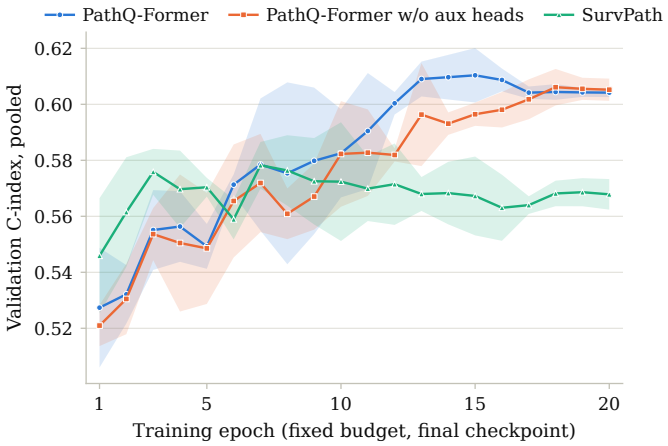


Fig. S1. Validation C-index per training epoch under the 20-epoch budget, pooled over all cohorts and folds, for PathQ-Former, its variant without auxiliary heads ('w/o aux heads') and SurvPath: mean over three seeds, ±1 sd band over seeds. All results use the final epoch; the curves only set the budget.

TABLE S4
SEED-AVERAGED PAIRED TESTS OF THE UNIMODAL BASELINES AGAINST SURVPATH OVER THE 25 POOLED (COHORT, FOLD) UNITS (DSS, 20 EPOCHS, THREE SEEDS): MEAN Δ C-INDEX, 95% BOOTSTRAP CI, t-TEST AND WILCOXON p, FOLDS WON. THE EIGHT REMAINING COMPARISONS ARE TABLE IV OF THE MAIN PAPER.

Comparison	Δ	95% CI	t p	W p	Wins
ABMIL vs SurvPath	+0.001	[−0.024, +0.026]	0.95	0.85	14/25
SNN vs SurvPath	−0.007	[−0.060, +0.045]	0.80	0.73	10/25
RNA MLP vs SurvPath	+0.024	[−0.027, +0.077]	0.38	0.41	15/25

repository) show SurvPath flat from about epoch 7 and both PathQ-Former variants rising until about epoch 13–18. The learning rate and gradient-accumulation setting of PathQ-Former were changed at the same time as the budget, so the 10- and 20-epoch results are not a pure budget comparison; a single-seed check at lr 2×10^{-4} on BLCA under the 20-epoch budget did not improve on the default (0.615 vs 0.630; its seed-averaged margin over SurvPath on the five BLCA folds was +0.020, 3 of 5 folds, against +0.036 for the default). The auxiliary-head variant and $\lambda = 0.5$ were adopted after the 20-epoch results of the variant without auxiliary heads on all five cohorts had been seen; the two variants are therefore reported as co-primary throughout, and no per-cohort result on BLCA is treated as confirmatory. Best-epoch validation scores were logged but used only in the selection replay (Table S11).

S-III. FULL TABLES

Table S4 completes the seed-averaged paired comparisons of Table IV of the main paper with the unimodal baselines against SurvPath: none differs from SurvPath pooled (every CI spans zero). Table S5 gives the per-cohort tests: on DSS, BRCA is the only cohort with a raw $p < 0.05$ and it does not survive Holm correction; on OS (one seed) BLCA, the development cohort, survives under the t-test only (Holm $p = 0.039$; Wilcoxon 0.312).

Table S6 reports the overall-survival endpoint (one seed): PathQ-Former leads SurvPath by 0.034 pooled, and the same checkpoint keeps a C-index of 0.531–0.586 with the RNA removed and 0.550–0.592 with the slide removed; with half of the validation patients missing one modality at random its OS C-index changes by at most 0.021 (BRCA, 0.596 to 0.576; BLCA, COADREAD and HNSC by 0.005, STAD by 0.012). Table S7 reports the 10-epoch budget: the variant without auxiliary heads leads SurvPath by 0.028 pooled (18 of 25 folds), COADREAD being the only cohort with a raw t-test $p < 0.05$ (not a pure budget comparison; Section S-II).

Table S8 reports the secondary metrics; the integrated Brier score (IBS, lower is better) is undefined on BRCA fold 4 for every method, so that fold is dropped from the paired units ($n = 24$). Uno's C-index tracks Harrell's: PathQ-Former leads SurvPath by 0.039 and the variant without auxiliary heads by 0.046. Each of the four other methods beats SurvPath on the IBS by about 0.07 (Wilcoxon $p < 10^{-4}$): SurvPath's hazards are poorly calibrated under this protocol (its published loss weights the censored term by 0.5), so the IBS is not a fair accuracy comparison and is reported for completeness only.

Table S9 gives the risk-score correlations behind Section IV-A of the main paper: the fused risk of PathQ-Former follows its own RNA-removed risk ($\rho = 0.83$ pooled) far more closely than its slide-removed risk (0.45), and the two single-input risks are nearly uncorrelated (0.15); the standard deviation over the 15 pairs of a cell is 0.06–0.24 and is in the archive. Table S10 gives the per-cohort values behind Section IV-D: mean imputation of a removed slide costs the variant without auxiliary heads 0.069 against 0.030

TABLE S5
PER-COHORT SEED-AVERAGED PAIRED DIFFERENCES AGAINST SURVPATH (FIVE FOLDS PER COHORT): RAW AND HOLM-ADJUSTED p-VALUES (t AND WILCOXON ADJUSTED SEPARATELY), FROM UNROUNDED VALUES. †DEVELOPMENT COHORT.

Cohort	Δ	t p (Holm)	W p (Holm)	Wins
PQF vs SurvPath (DSS, three seeds)				
BLCA†	+0.036	0.175 (0.700)	0.312 (1.000)	3/5
BRCA	+0.086	0.017 (0.084)	0.062 (0.312)	5/5
COADREAD	+0.044	0.583 (1.000)	0.625 (1.000)	4/5
HNSC	+0.031	0.544 (1.000)	0.625 (1.000)	4/5
STAD	−0.015	0.719 (1.000)	0.625 (1.000)	2/5
PQF⁻ vs SurvPath (DSS, three seeds)				
BLCA†	+0.015	0.518 (1.000)	0.625 (1.000)	3/5
BRCA	+0.075	0.073 (0.367)	0.125 (0.625)	4/5
COADREAD	+0.068	0.202 (0.807)	0.188 (0.750)	4/5
HNSC	+0.023	0.580 (1.000)	0.625 (1.000)	4/5
STAD	+0.006	0.847 (1.000)	0.812 (1.000)	2/5
PQF vs SurvPath (OS, seed 0)				
BLCA†	+0.043	0.008 (0.039)	0.062 (0.312)	5/5
BRCA	+0.073	0.015 (0.058)	0.062 (0.312)	5/5
COADREAD	−0.015	0.578 (0.578)	0.625 (0.625)	2/5
HNSC	+0.025	0.046 (0.138)	0.125 (0.375)	4/5
STAD	+0.044	0.230 (0.460)	0.312 (0.625)	3/5

TABLE S6
OVERALL SURVIVAL (20 EPOCHS, SEED 0): FIVE-FOLD MEAN C-INDEX ± SD OVER FOLDS, AND THE PATHQ-FORMER CHECKPOINT WITH ONE INPUT REMOVED (NULL CODES). †DEVELOPMENT COHORT.

Cohort	PQF	SurvPath	Δ	RNA rem.	Slide rem.
BLCA†	0.595 ± 0.053	0.552 ± 0.056	+0.043	0.576	0.592
BRCA	0.596 ± 0.043	0.523 ± 0.052	+0.073	0.531	0.556
COADREAD	0.581 ± 0.132	0.596 ± 0.153	−0.015	0.586	0.562
HNSC	0.536 ± 0.036	0.510 ± 0.029	+0.025	0.535	0.550
STAD	0.603 ± 0.070	0.559 ± 0.045	+0.044	0.574	0.582
Pooled	0.582	0.548	+0.034		

TABLE S7
10-EPOCH BUDGET (DSS, THREE SEEDS): C-INDEX MEAN ± SD OVER SEEDS OF THE FIVE-FOLD MEAN, AND THE SEED-AVERAGED PAIRED DIFFERENCE (POOLED: 25 FOLDS, 18 WON). †DEVELOPMENT COHORT.

Cohort	PQF⁻	SurvPath	Δ	t p	W p
BLCA†	0.618 ± 0.014	0.589 ± 0.004	+0.029	0.272	0.312
BRCA	0.600 ± 0.031	0.536 ± 0.015	+0.064	0.142	0.188
COADREAD	0.646 ± 0.046	0.551 ± 0.022	+0.094	0.015	0.062
HNSC	0.556 ± 0.007	0.568 ± 0.035	−0.013	0.595	1.000
STAD	0.574 ± 0.025	0.607 ± 0.012	−0.033	0.471	0.625
Pooled	0.599	0.570	+0.028	0.079	0.063

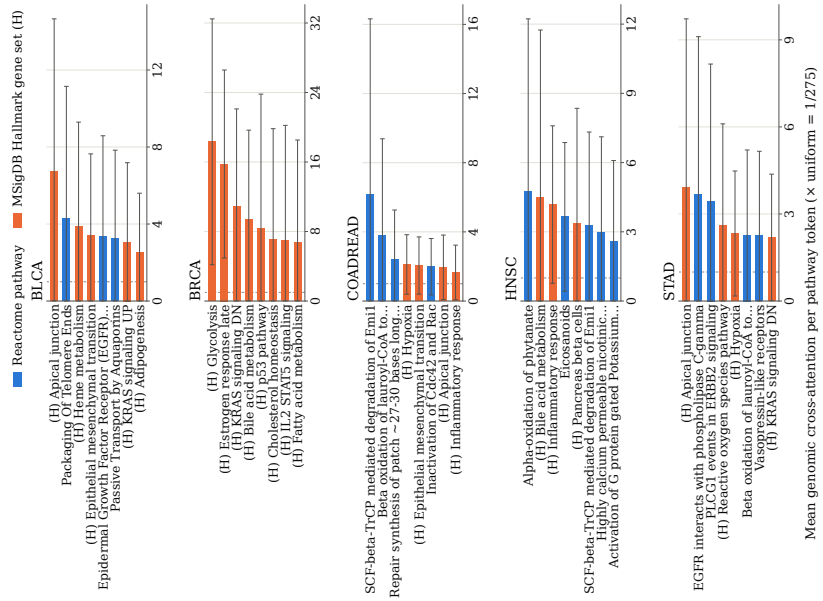


Fig. S2. Pathway tokens with the highest mean genomic-query attention per cohort (PathQ-Former, 20 epochs, seed 0), as a multiple of uniform attention over the 275 tokens (dashed line), averaged over validation patients and the five folds (rotated). Blue: Reactome; orange, (H): MSigDB Hallmark. Error bars: mean ± sd over the five folds, clipped at zero; names over 45 characters are shortened.

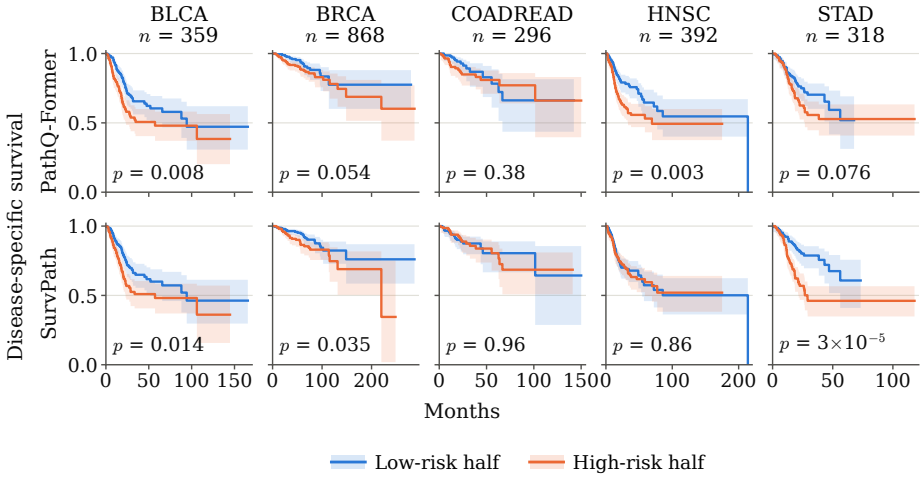


Fig. S3. Kaplan–Meier curves of disease-specific survival for the high- and low-risk halves of the pooled out-of-fold validation predictions (20 epochs, seed 0); top: PathQ-Former, bottom: SurvPath. Patients are split at the median predicted risk of their own validation fold and the folds pooled, so every patient appears once. Bands: 95% CI; p: log-rank test between the halves.

with the null code, and PathQ-Former 0.035 against 0.025 (null-code rows: Table II of the main paper).

Table S12 gives the fusion-block ablations on BLCA under the 10-epoch budget (Section S-II); the settings move the fused C-index within 0.595–0.638, inside the fold standard deviation of 0.07–0.13, and are not distinguishable at one seed. Table S13 ablates the final configuration under the 20-epoch budget on BLCA (one seed each, trained on a laptop CPU) against the seed-0 GPU run of the main campaign (0.625 with both inputs; 0.630 ± 0.010 over three seeds). Capping PathQ-Former at the 4,096 training patches per slide that SurvPath uses gives 0.626, so on this cohort the margin over SurvPath is not explained by PathQ-Former reading more patches; two time bins give 0.628 and eight 0.609; the 281 Xena pathway groups [56] give 0.628 and the 50 Hallmark gene sets alone 0.600, 0.025 below the 275-token Reactome and Hallmark set. With half of the validation patients missing one modality at random every variant stays within 0.015 of its complete-input score (10–30%: in the archive); for the two-bin variant the RNA- and slide-permuted evaluations give 0.619 and 0.571. All differences are within one fold standard deviation (0.057–0.082) and single-seed.

Table S11 is the checkpoint-selection replay summarised in Section IV-E of the main paper. Block (A) replays three post-hoc rules on the stored 20-epoch histories (three seeds; the single-modality PathQ-Formers have 70 instead of 75 fold histories): validation-loss early stopping (patience 5) costs the six PathQ-Former, SNN and MLP runs 0.027–0.090 pooled and ABMIL and SurvPath nothing, because the latter reach their best validation C-index at epoch 6.2 and 6.9 against 7.2–10.0 for the others; the oracle rule, an unattainable bound biased in favour of every method, puts every method at 0.67–0.69. The two 10-epoch methods behave the same way (early stopping costs PQF$^-$ 0.075 and SurvPath 0.010). Block (B) is the earlier campaign that actually trained with validation-loss selection (patience 5, minimum 6 epochs, 20-epoch budget; 29 cohort-level runs, 145 folds; Section S-VIII): its reported C-index is 0.063–0.128 below the best epoch of the

TABLE S8
SECONDARY METRICS (20-EPOCH DSS PROTOCOL): MEAN ± SD OVER THREE SEEDS OF THE FIVE-FOLD MEAN, AND THE SEED-AVERAGED PAIRED DIFFERENCE FROM SURVPATH OVER THE POOLED FOLDS (95% BOOTSTRAP CI; t-TEST / WILCOXON p). *FOUR FOLDS (UNDEFINED ON BRCA FOLD 4).

	PQF	PQF$^-$	ABMIL	RNA MLP	SurvPath
Uno's IPCW C-index					
BLCA	0.630 ± 0.009	0.612 ± 0.019	0.560 ± 0.014	0.605 ± 0.013	0.597 ± 0.010
BRCA	0.636 ± 0.012	0.648 ± 0.038	0.599 ± 0.012	0.598 ± 0.012	0.525 ± 0.049
COADREAD	0.616 ± 0.032	0.644 ± 0.009	0.629 ± 0.015	0.646 ± 0.025	0.592 ± 0.020
HNSC	0.589 ± 0.034	0.587 ± 0.012	0.563 ± 0.031	0.563 ± 0.013	0.556 ± 0.027
STAD	0.586 ± 0.019	0.601 ± 0.014	0.562 ± 0.029	0.550 ± 0.010	0.591 ± 0.011
Δ vs SurvPath ($n = 25$), 95% CI	$+0.039$ $[-0.016, +0.086]$	$+0.046$ $[+0.004, +0.087]$	$+0.010$ $[-0.019, +0.042]$	$+0.020$ $[-0.031, +0.070]$	
t p / W p	0.153 / 0.042	0.045 / 0.042	0.526 / 0.751	0.455 / 0.474	
Integrated Brier score					
BLCA	0.205 ± 0.008	0.210 ± 0.004	0.202 ± 0.007	0.191 ± 0.002	0.268 ± 0.031
BRCA*	0.073 ± 0.000	0.082 ± 0.004	0.077 ± 0.002	0.075 ± 0.002	0.131 ± 0.005
COADREAD	0.077 ± 0.006	0.081 ± 0.003	0.097 ± 0.001	0.082 ± 0.002	0.110 ± 0.009
HNSC	0.221 ± 0.019	0.215 ± 0.011	0.205 ± 0.010	0.200 ± 0.005	0.341 ± 0.087
STAD	0.201 ± 0.006	0.189 ± 0.009	0.208 ± 0.013	0.204 ± 0.001	0.276 ± 0.043
Δ vs SurvPath ($n = 24$), 95% CI	-0.070 $[-0.100, -0.044]$	-0.071 $[-0.101, -0.043]$	-0.068 $[-0.100, -0.039]$	-0.075 $[-0.108, -0.044]$	
t p / W p	<0.001 / <0.001	<0.001 / <0.001	<0.001 / <0.001	<0.001 / <0.001	

TABLE S9
RISK-SCORE CORRELATIONS OF PATHQ-FORMER (20 EPOCHS): SPEARMAN ρ, MEAN OVER THE 15 (FOLD, SEED) PAIRS PER COHORT, WITH ITS OWN SINGLE-INPUT RISKS (TOP) AND THE OTHER METHODS' RISKS ON THE SAME FOLD AND SEED (BOTTOM); POOLED: MEAN OVER THE 75 PAIRS.

Fused risk of PQF vs	BLCA	BRCA	COADREAD	HNSC	STAD	Pooled
own slide-removed risk	0.42	0.36	0.53	0.47	0.45	0.45
own RNA-removed risk	0.90	0.85	0.73	0.84	0.82	0.83
(slide-rem. vs RNA-rem.)	0.22	0.09	0.16	0.15	0.13	0.15
PQF$^-$	0.79	0.62	0.66	0.71	0.69	0.69
SurvPath	0.63	0.38	0.38	0.45	0.48	0.46
ABMIL	0.46	0.51	0.38	0.44	0.48	0.46
RNA MLP	0.42	0.26	0.37	0.29	0.41	0.35
SNN	0.38	0.23	0.36	0.29	0.43	0.34

same runs for every method. Selection and scoring use the same fold throughout, so every replayed rule is biased in its own favour and the oracle is an upper bound. (For the single-modality PathQ-Formers the run-weighted Final differs by 0.001–0.002 from the Pooled column of Table III of the main paper, which averages the cohort means.)

S-IV. EFFICIENCY

Table S14 reports parameters, latency and peak GPU memory per patient (first 40 patients of the BLCA fold-0 validation set; medians on one RTX PRO 4000). At 4,096–6,214 patches the two models have equal forward latency; PathQ-Former's training step takes 104 ms against 176 ms for SurvPath, and its peak memory on full slides is 1.17 GiB against 1.74 GiB. The claim is limited to these slide sizes on this GPU; neither implementation was profiled, and training throughput in practice was bound by reading features from storage.

S-V. PATHWAY ATTENTION

The genomic query block exposes cross-attention from each query over the 275 pathway tokens. Fig. S2 shows, per cohort, the eight tokens with the highest mean attention relative to uniform ($1/275$), averaged over validation patients and the five folds (seed 0). The top entries are 3.9 to 18.3 times the uniform weight, but the standard deviation over folds exceeds the mean for most of them (BLCA: apical junction, 6.7 ± 7.9; BRCA: glycolysis, 18.3 ± 14.1), so the ranking is descriptive only; attention weights are not a measure of feature importance [51].

S-VI. KAPLAN–MEIER CURVES

Fig. S3 shows disease-specific survival of the high- and low-risk halves of the pooled out-of-fold predictions (20 epochs, seed 0). The log-rank p-values between the halves are, for PathQ-Former, 0.008 (BLCA), 0.054 (BRCA), 0.38 (COADREAD), 0.003 (HNSC) and 0.076 (STAD), and for SurvPath 0.014, 0.035, 0.96, 0.86 and 3×10^{-5}. The pooled curves combine five fold models at one seed and are descriptive.

S-VII. EVALUATION PITFALLS IN SURVIVAL MIL PIPELINES

Each of the following choices produces plausible-looking C-index values in a survival MIL pipeline; ours avoids them as stated. (1) *Concordance on discrete bins* ties many pairs; Harrell's C-index is computed on continuous time. (2) *Bin edges fit on the validation fold* put its labels on a different scale and leak its label distribution; edges are fit on uncensored training patients and reused. (3) *Slide-level samples* weight multi-slide patients several times; one sample per patient, all slides concatenated. (4) *Unscaled gene inputs* feed log-expression on an arbitrary scale to the pathway MLPs; genes are min–max scaled with training-split statistics. (5) *Selection on the reported fold* biases every rule upward (Table S11); a fixed budget, the final checkpoint and several seeds avoid it. (6) *Non-standard risk and dropout semantics*, $1 - S(T)$ as the risk or modality dropout per batch; risk is $-\sum_t S(t)$ and dropout is per sample. (7) *Extreme times in the Brier grid* evaluate where a single patient is at risk; the interior quartile edges are used. (8) *Untruncated IPCW weights* diverge as the censoring survival function approaches zero; Uno's C-index is truncated at the 75th percentile of training event times.

S-VIII. RUN INVENTORY AND COMPUTE

The archive holds 204 cohort-level GPU runs, each of five folds, and 13 cohort-level runs trained on a laptop CPU. The GPU runs are: 34 in the first multi-cohort campaign, namely 24 under the earlier validation-loss protocol and the single-seed 10-epoch runs of PathQ-Former and SurvPath on five cohorts; 31 BLCA fusion-block ablations, late fusion and 10-epoch seeds 1–2 of PathQ-Former and SurvPath; 128 under the 20-epoch DSS protocol (the nine methods of Table S2 on five cohorts with the seeds listed there); 10 under the OS endpoint (PathQ-Former and SurvPath on five cohorts); and one 20-epoch learning-rate check on BLCA (Section S-II). The laptop runs are the eight BLCA seed-0 runs of the validation-loss campaign and the five final-configuration ablations of Table S13. The validation-loss campaign (patience 5, minimum 6 epochs, 20-epoch budget) comprises 29 cohort-level runs with 145 folds: SurvPath, ABMIL, SNN, the RNA MLP and the earlier fusion configuration on five cohorts, with three seeds on BLCA for SurvPath and the fusion configuration (Table S11B; three further BLCA PathQ-Former runs,

TABLE S10
MEAN IMPUTATION ON THE SAME PATHQ-FORMER CHECKPOINTS (20 EPOCHS): C-INDEX WITH ONE INPUT REPLACED BY THE TRAINING-MEAN PATCH EMBEDDING OR GENE VECTOR (MEAN ± SD OVER THREE SEEDS OF THE FIVE-FOLD MEAN; ONE SEED FOR PQF$^-$), AND THE POOLED DIFFERENCE FROM THE COMPLETE-INPUT SCORE OVER THE 25 SEED-AVERAGED FOLDS.

Condition	BLCA	BRCA	COADREAD	HNSC	STAD	Δ	t p	W p
PQF, mean imputation (three seeds)								
RNA removed	0.597 ± 0.003	0.607 ± 0.035	0.602 ± 0.012	0.589 ± 0.014	0.581 ± 0.018	-0.009	0.29	0.49
Slide removed	0.595 ± 0.027	0.591 ± 0.060	0.580 ± 0.016	0.558 ± 0.024	0.521 ± 0.037	-0.035	0.040	0.045
PQF$^-$, mean imputation (one seed)								
RNA removed	0.627	0.545	0.631	0.580	0.594	-0.010	0.50	0.83
Slide removed	0.524	0.499	0.600	0.535	0.524	-0.069	3×10^{-4}	2.5×10^{-4}

TABLE S11
CHECKPOINT SELECTION REPLAYED ON THE STORED VALIDATION HISTORIES. (A) FINAL: FIXED-BUDGET FINAL CHECKPOINT; EARLY: VALIDATION-LOSS EARLY STOPPING, PATIENCE 5; MIN. LOSS: LOWEST VALIDATION LOSS; ORACLE: HIGHEST VALIDATION C-INDEX (POOLED MEANS OVER (FOLD, SEED) RUNS); Δ: EARLY MINUS FINAL OVER THE 25 SEED-AVERAGED FOLDS; SEL. / BEST: MEAN SELECTED EPOCH / MEAN EPOCH OF HIGHEST VALIDATION C-INDEX. (B) THE EARLIER VALIDATION-LOSS CAMPAIGN: REPORTED C-INDEX, C-INDEX AT THE BEST AND THE FINAL EPOCH TRAINED, MEAN SELECTED EPOCH / EPOCHS TRAINED.

(A) Replay on the stored histories

Method	Budget	Folds	Final	Early	Min. loss	Oracle	Δ	t p	W p	Sel. / Best
PQF	20	75	0.604	0.546	0.558	0.689	-0.059	0.016	8×10^{-4}	2.2 / 10.0
PQF$^-$	20	75	0.605	0.516	0.527	0.687	-0.090	3×10^{-7}	1×10^{-6}	1.8 / 9.9
Slide-only PathQ-Former	20	70	0.588	0.523	0.529	0.688	-0.065	0.006	0.005	1.8 / 9.3
RNA-only PathQ-Former	20	70	0.567	0.532	0.539	0.681	-0.034	0.062	0.080	1.4 / 9.7
SurvPath (official)	20	75	0.568	0.568	0.559	0.679	$+0.000$	0.99	0.98	1.6 / 6.9
ABMIL	20	75	0.569	0.568	0.568	0.668	-0.001	0.95	0.61	1.4 / 6.2
SNN	20	75	0.561	0.523	0.522	0.669	-0.038	0.054	0.052	1.4 / 7.3
RNA MLP	20	75	0.592	0.565	0.565	0.679	-0.027	0.078	0.075	1.2 / 7.2
PQF$^-$	10	75	0.599	0.524	0.538	0.658	-0.075	8×10^{-6}	2.7×10^{-5}	1.8 / 6.3
SurvPath (official)	10	75	0.570	0.560	0.560	0.663	-0.010	0.55	0.47	1.4 / 4.5

(B) The earlier validation-loss campaign (selection actually applied in training)

Method	Budget	Folds	Reported	Best epoch	Final epoch	Sel. / Trained
SurvPath (official)	20	35	0.562	0.642	0.549	1.3 / 6.4
ABMIL	20	25	0.535	0.639	0.568	1.5 / 6.5
SNN	20	25	0.514	0.642	0.561	1.3 / 6.4
RNA MLP	20	25	0.545	0.659	0.578	1.2 / 6.3
Earlier fusion configuration	20	35	0.577	0.641	0.569	4.7 / 9.8

TABLE S12
FUSION-BLOCK ABLATIONS ON BLCA, 10-EPOCH BUDGET (PQF$^-$, SEED 0): FIVE-FOLD MEAN C-INDEX ± SD OVER FOLDS, AND WITH ONE INPUT REMOVED (NULL CODES).

Variant	Both inputs	RNA rem.	Slide rem.
Base (K=32, depth 2, mod. dropout 0.15)	0.609 ± 0.069	0.602	0.630
Modality dropout 0	0.634 ± 0.075	0.609	0.598
Modality dropout 0.3	0.616 ± 0.068	0.624	0.648
Modality dropout 0.5	0.635 ± 0.108	0.587	0.626
$K = 16$ queries	0.638 ± 0.070	0.615	0.615
$K = 64$ queries	0.634 ± 0.127	0.619	0.624
Fusion depth 1	0.598 ± 0.124	0.584	0.611
Fusion depth 3	0.595 ± 0.088	0.585	0.610

TABLE S13
FINAL-CONFIGURATION ABLATIONS ON BLCA, 20-EPOCH BUDGET (PQF, SEED 0, LAPTOP CPU): FIVE-FOLD MEAN C-INDEX ± SD, WITH ONE INPUT REMOVED (NULL CODES) AND WITH 50% OF VALIDATION PATIENTS MISSING ONE MODALITY AT RANDOM (FIVE DRAWS PER FOLD). BASE: SEED-0 GPU RUN (4 BINS, 275 TOKENS, ALL PATCHES).

Variant	Both inputs	RNA rem.	Slide rem.	50% missing
Base (seed 0, GPU)	0.625 ± 0.081	0.598	0.588	0.618
Time bins $T = 2$	0.628 ± 0.057	0.634	0.575	0.616
Time bins $T = 8$	0.609 ± 0.082	0.578	0.605	0.604
Hallmark only (50 tokens)	0.600 ± 0.075	0.608	0.554	0.597
Xena groups (281 tokens)	0.628 ± 0.063	0.604	0.634	0.615
4,096 patches per slide	0.626 ± 0.059	0.591	0.622	0.611

one without warm-up and the slide-only and RNA-only variants of the earlier fusion configuration, are archived but not tabulated). It selected epoch 1 on median (mean 2.2; 82% of folds epoch 1–3), trained 7.2 epochs on average, and reported a mean C-index of 0.543 against 0.644 at the best epoch of the same runs, with 14 of 25 method × cohort cells within 0.45–0.55; at or below 0.55 the counts are STAD 4 of 5, HNSC 5 of 5 and BRCA 4 of 5, the exceptions being SurvPath on STAD (0.610) and SNN on BRCA (0.572). Replaying each run's own rule on its stored history reproduces the stored selected epoch and reported C-index in 145 of 145 folds.

TABLE S14
COMPUTE PER PATIENT (BLCA FOLD-0 VALIDATION, FIRST 40 PATIENTS, MEDIANS, ONE RTX PRO 4000): PARAMETERS (M), PATCHES (6,214 = MEDIAN PATIENT), FORWARD AND FORWARD+BACKWARD TIME (MS), PEAK GPU MEMORY (GIB).

Model	Params	Patches	Fwd	Fwd+bwd	Mem.
PQF (all patches)	16.9	6,214	35.0	104.2	1.17
PQF (4,096 patches)	16.9	4,096	34.7	104.1	0.17
SurvPath (as trained)	21.2	4,096	35.7	175.6	0.20
SurvPath (all patches)	21.2	6,214	36.0	176.0	1.74

Across the 128 runs of the 20-epoch DSS protocol, the standard deviation of the C-index over the five folds of a run ranges from 0.03 (SNN on BRCA at seed 1; both PathQ-Former variants on HNSC at seed 0) to 0.25 (COADREAD, PathQ-Former at seed 1); on COADREAD it is 0.12–0.25 for the fusion models and 0.05–0.25 over all methods, which is why per-cohort tests with five folds are exploratory. Every run stores its results, configuration, per-fold predictions and per-epoch histories (late-fusion runs: results and predictions only); GPU runs used rented RTX PRO 4000/4500 and RTX 2000 Ada GPUs sharing a network volume.